

MAHMOUD SALAMA

Enterprise technology, transformation and architecture at scale.

A curated executive portfolio spanning national procurement and supply-chain transformation, enterprise architecture, platform modernization, data and integration, security and resilience, government workflow, education, digital health and 150+ major and mid-sized delivery engagements across Egypt and Oman.

18+ years	54 technology organization	7 direct reports	EGP 150M annual IT budget
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MedIQ

The defining enterprise program in my current career chapter: a national governmental procurement and supply-chain platform connecting procurement, supply, inventory, health entities, suppliers, enterprise integrations, governed workflows, data and executive visibility at national scale.

38K+ health-team users	11K+ healthcare facilities	2K+ suppliers	~6 months end-to-end core replacement
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Current leadership & ownership

At UPA, I lead a 54-person technology organization with seven direct reports and final technical/architecture decision ownership. My scope includes technology strategy, Enterprise/Solution Architecture, platform modernization, SDLC and release governance, data and integration governance, security and business continuity, vendor governance, infrastructure, service operations and controlled enterprise AI adoption.

MedIQ core modernization - winner of the Pharma Awards 2026 Digital Transformation Award

- Full re-architecture and replacement rather than a simple upgrade or refactor.
- Approximately three months for the core rebuild and approximately six months end-to-end including testing, security/performance validation, migration and reconciliation, training/readiness and cutover.
- Gradual migration followed by a Big-Bang cutover after readiness, with approximately one month of parallel old/new operation.
- 100% in-house delivery using the full technology organization.
- Current technology evidence includes ReactJS/Node.js, C#/.NET APIs, PostgreSQL, Redis, Docker, reporting platforms and enterprise APIs.

Evidence discipline: period-specific scale metrics are kept separate. FY2024/25 evidence records 35,338 users, 5,991 requesting government entities, 2,155 suppliers, 89,761 warehouses/stores, 35,357 purchase orders and 938,242 recipient invoices. A separate 2020–2025 operational reference records 633,896 supply requests and 160,538 supply orders.

National platforms & governed workflows.

FLAGSHIP · UPA · WORKFLOW GOVERNANCE MedIQ E-Tender End-to-end electronic tender/practice platform covering 10 linked steps, 30 operational stages, 15 Chairman approvals and 20+ governed roles/permissions, with segregation of duties, workflow gates, traceability and controlled award-to-framework-agreement outputs. Enterprise workflow, procurement-governance and architecture leadership	FLAGSHIP · UPA · MOBILE Mobile MedIQ / Electronic Inventory Mobile-enabled field stocktaking and physical-to-system reconciliation with digital stocktake records. Latest evidence includes 20,513 app downloads, 17,236 workshop attendees and 13,226 inventory committees. Platform, delivery and operational-governance leadership
UPA · NATIONAL INVENTORY National Inventory Governance Governance model across ~89,000 registered warehouses/stores, defining strategic, main, regional and subordinate roles with request/dispensing controls. A June 2026 baseline across three health sectors covered 18,875 stores: 79% active, 18.2% active without consumption and 2.8% inactive, guiding intervention priorities. Technology, operating-model and governance leadership	FLAGSHIP · UPA · MEDICAL ASSETS Unified Assets & Maintenance Digitized medical-device inventory and unified maintenance visibility linked to MedIQ, including MapTiler-based geographic mapping and GS1 GTIN/GLN integration. Evidence covers 242,816 devices across 164 entities; the program also shifted maintenance from paper toward governed electronic workflows with legal/process alignment and user adoption. Architecture, decision ownership and transformation leadership

Recent operational evidence

For 1-18 August 2026, electronic inventory operations recorded 332,712 consumption transactions, EGP 23.79B transaction value, 6,503 warehouses/stores and 783 entities.

Integration, capability architecture, data, AI & resilience.

UPA · ENTERPRISE INTEGRATION Microsoft Dynamics 365 Enterprise Integration Owned the integration study, architecture and decision model for MediQ ↔ Dynamics 365 across master data, procurement, supply, inventory, finance and BI using APIs, middleware/integration services and governed DB views. Directed the completed EGP 11M delivery through SIT/UAT, controlled deployment and stabilization. Integration architecture, decision ownership and delivery governance	UPA · DATA / AI Enterprise Data & AI Architecture Defined and owned the target five-layer data architecture spanning operational sources, validation/cleansing, analytical models, reporting/dashboards and phased AI. The roadmap covers demand forecasting, shortage and expiry risk, unusual-order detection, redistribution, supplier performance, anomaly detection and scenario simulation. Data architecture, governance and controlled AI strategy
UPA · ENTERPRISE CAPABILITIES Enterprise Capability Architecture Defined target-state capability architecture for a connected MediQ ecosystem spanning supply, tenders, inventory, finance, assets, maintenance, e-pharmacy, complaints/customer service, content and connected healthcare services, aligning platform boundaries and transformation sequencing. Enterprise architecture and target-state decision ownership	UPA · CUSTOMER EXPERIENCE Customer Experience & Technical Support Transformation Owned the transformation study and target operating-model decision for CRM + IVR + Ticketing, evaluating Dynamics/Odoo, specialist outsourcing and a hybrid transition across data control, time-to-market, customization, CapEx/OpEx, resilience, scalability, knowledge retention and integration. Build-vs-buy, TCO and operating-model decision ownership
UPA · CROSS-GOVERNMENT INTEGRATION Healthcare & Government Interoperability Architected and governed integrations connecting MediQ with ECIC, MoH/HMIS, GAHAC and Blood Bank systems; ECIC and Blood Bank are confirmed production integrations. Separately, personally designed and developed production REST APIs for secure beneficiary, prescription and dispensing-data exchange. Enterprise interoperability architecture and production-delivery ownership	UPA · SECURITY & RESILIENCE Security, Resilience & Production Operations Defense-in-depth with IAM/RBAC, least privilege, segregation of duties, MFA/OTP, TLS/API controls, F5 BIG-IP, network segmentation, container security, monitoring/logging, PostgreSQL replication, backup/recovery and a DR environment. Architecture and technical decision ownership

Architecture decision model

Requirements and constraints → alternatives → PoC/prototype when needed → decision with the relevant technical team → implementation → testing and performance/security validation → controlled production. Architecture risk assessment covers scalability, single points of failure, security, migration, vendor dependency, technology dependency and lock-in.

Strategy, governance & operating resilience.

STRATEGY Technology Strategy & Roadmaps Portfolio-level current/target-state view across applications, capabilities, integrations, sources of truth, duplication, replacement/retirement and roadmap sequencing, tied to business priorities, investment, risk and value realization. Executive technology strategy ownership	ARCHITECTURE & DELIVERY Architecture & Delivery Governance Own architecture/design decisions, technical advisory, vendor/TOR evaluation, production readiness and controlled delivery. Scope also includes end-to-end logistics/customs workflow architecture and a centralized reusable reporting architecture with 20+ deployed reports. Enterprise architecture, advisory and delivery governance
DATA & INTEROPERABILITY Data & Integration Governance Govern API contracts, mapping, authentication, direction of exchange, reconciliation, exception handling, versioning and data ownership, with formal quality controls across completeness, validity, consistency, timeliness, uniqueness, traceability and matchability. Enterprise interoperability and data-governance ownership	INVESTMENT GOVERNANCE Applied Build vs Buy / TCO Technology and operating-model decisions consider licensing, infrastructure, implementation, support, staffing, resilience, scalability and knowledge retention. The CRM/IVR/Ticketing study provides a concrete applied case comparing on-prem, outsourcing and hybrid paths. Technology investment and operating-model decision ownership

Selected governance appointments

2025-2026 Egypt Digital Transformation Committee Egypt-side digital-transformation committee contribution supporting the World Bank Group B-READY Assessment, with technical input on digitization, modernization and national competitiveness. National-level technology and transformation contribution	2021-2024 National Blood Operations Oversight Board-level national oversight contribution relating to digital systems, data integrity and technology standards in the blood-supply environment. Board Member	2021-2024 Pharmaceutical Track & Trace National governance contribution supporting vision, compliance, interoperability, data integrity and technology-roadmap discussions across UPA, EDA and Ministry of Health stakeholders. Governance contribution
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Oman government & education platforms.

Representative work from a hands-on development and technical-leadership chapter in Oman. Client ministries and agencies are project contexts, not separate employers.

INTEGRAL · NATIONAL SCALE Oman National Educational Portal National education platform with hands-on contribution across analysis, applications, databases, reporting, testing, deployment, rollout and support; approximately 1M users in prior career evidence. Senior hands-on developer and key technical contributor	INTEGRAL · GOVERNMENT WORKFLOW Maktabi Electronic Correspondence Large correspondence system covering creation, routing, forwarding, tracking, reporting, database components, testing, release and support. Major hands-on developer within the delivery team
INTEGRAL · AWARD-WINNING Recruitment System Complete government recruitment solution from analysis and design through database, development, testing and production deployment; the solution received Sultan Qaboos Award recognition. Technical owner	INTEGRAL · EDUCATION Student Incident Management System for recording, following up and reporting student incidents across the education environment. Technical owner
INTEGRAL · EDUCATION International Schools Management Requirements, application and database development, testing, deployment and support. Core technical contributor	INTEGRAL · IT OPERATIONS IT Asset & Hardware Management Hardware-focused inventory and operational control for the Ministry of Education environment. Core technical contributor
INTEGRAL · AGRICULTURE Ministry of Agriculture Portal & Omani Dates Market Portal plus purchasing and request workflows for the Ministry of Agriculture environment. Technical owner	INTEGRAL · MEMBERSHIP & EVENTS Al Wahat Club Management Memberships and affiliates, facility reservations, approvals and event management for a Royal Court Affairs club assignment. Core technical contributor · client project

Reusable products, government portals & digital health.

KEYFRAME · PRODUCT FOUNDATION Reusable .NET CMS Multilingual/RTL publishing, pages, articles, news, events, media galleries, permissions, workflows, search, SEO, analytics, resource files and integrations. Sole application architect and developer	KEYFRAME · HEALTH Egypt Ministry of Health & Population Portal Large portal with services, CMS, electronic library, FAQ, newsletter, polls, media, social integration, reporting, security and deployment. Technical owner
KEYFRAME · DIGITAL GOVERNMENT Egypt Ministry of Planning Portal Analysis, SRS authorship, application architecture, database, CMS, development, testing and deployment within the Keyframe client engagement. Technical owner and SRS author	KEYFRAME · GOVERNMENT SUITE Egypt Government Digital Portals Representative work includes MSAD, Rate Your Services, NMI, the national anti-corruption committee portal, Egypt Calendar, Elections in Egypt and surveys. Technical owner · client assignments
KEYFRAME · CONFERENCE PLATFORM Arab Medical Union, UCHID, BGICC & Arab Organizers Registration, attendance, sessions, speakers, research evaluation, certificates, communications, administration and reporting. Sole application architect and developer	KEYFRAME · ITIDA RECOGNITION Yashfeen Medical Portal Full medical portal recognized through an ITIDA award during the Keyframe period. Technical owner · award-recognized solution
KEYFRAME · PUBLIC SAFETY Traffic Reporting & Officers' Affairs Complete solution for traffic reports and officers' affairs, from analysis and database design through testing and deployment. Technical owner	KEYFRAME · EDUCATION Electronic Examinations & Review Electronic examinations and curriculum-review system for multiple educational certificate curricula. Technical owner
KEYFRAME · MULTI-COUNTRY Electrolux Websites Reusable CMS customized for Electrolux in Egypt, Saudi Arabia, the UAE and Turkey. Technical owner	INDEPENDENT VENTURE · DIGITAL HEALTH Salamat Healthcare Platform Bilingual healthcare platform developed as an independent digital-health product venture during the Keyframe period. Founder and sole product/application developer

Representative, defensible work selected to demonstrate scale, architecture, transformation and delivery depth. Smaller engagements and confidential identifiers are deliberately omitted.